

Deep Learning Algorithms for Landslide Detection using Synthetic Aperture Radar Data

Thesis

Submitted in partial fulfilment of the requirements for the degree of

MASTER OF TECHNOLOGY

in

Signal Processing and Machine Learning

by

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NATIONAL INSTITUTE OF TECHNOLOGY, KARNATAKA
SURATHKAL, MANGALORE - 575025

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Under the guidance of

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DECLARATION

by the P.G (M.Tech) Student

I hereby declare that the report of the P.G. Project Work entitled **Deep Learning Algorithms for Landslide Detection using Synthetic Aperture Radar Data.** which is being submitted to the National Institute of Technology Karnataka, Surathkal, in partial fulfilment of the requirements for the award of the Degree of **Master of Technology in Signal Processing and Machine Learning** in the Department of Electronics and Communication Engineering, is a bonafide report of the work carried out by me. The material contained in this report has not been submitted to any University or Institution for the award of any degree.

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Department of Electronics and Communication Engineering

Place: NITK, Surathkal

Date: July 21, 2025

CERTIFICATE

This is to certify that the P.G Project Work Report entitled **Deep Learning Algorithms for Landslide Detection using Synthetic Aperture Radar Data.** submitted by **Manav Rajpal, (Register No. 2320207)** as the record of the work carried out by him, is accepted as the P.G Project Work Report submission in partial fulfillment of the requirements for the award of degree of **Master of Technology in Signal Processing and Machine Learning** in the Department of Electronics and Communication Engineering, National Institute of Technology Karnataka, Surathkal, during the academic year 2021-2023.

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ABSTRACT

Landslides are a significant geohazard, often triggered by seismic or extreme weather events. Timely and accurate detection is critical for effective risk mitigation and disaster response. Synthetic Aperture Radar (SAR) plays a key role in this, thanks to its ability to capture images in all weather conditions, day or night.

This study explores the use of deep learning algorithms for landslide segmentation using SAR data. In this study, advanced semantic segmentation models, including UNet, Linknet, UNet++, PAN, DeepLab-v3+, DRs-UNet, MSSCSAF-Net and ASKU-Net++ are implemented and evaluated on a high-resolution dataset from the 2017 Hokkaido earthquake.

The model's performance was assessed using metrics like Precision, Recall, F1 Score, IoU and AUPRC. Among all models tested, DeepLabv3+, ASKU-Net++ and MSSCSAF-Net achieved the best results, with the later consistently securing the highest F1 scores across different sampling strategies and regions. These results underscore the importance of integrating multi-scale context, spatial attention, and terrain-specific features in developing robust SAR-based landslide detection systems.

Keywords: Landslide detection, SAR, Deep learning, Semantic segmentation

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ABBREVIATIONS

AUPRC	Area Under the Precision-Recall Curve
CNN	Convolutional Neural Network
DEM	Digital Elevation Model
PAN	Pyramid Attention Network
SAR	Synthetic Aperture Radar
VV	Vertical Transmit and Vertical Receive Polarization
VH	Vertical Transmit and Horizontal Receive Polarization
TP	True Positive
FP	False Positive
FN	False Negative

Chapter 1

Introduction

Landslides are recognized as among the most destructive and horrifying of all natural hazards that may occur to an area, with extensive loss of life, large destruction of infrastructure, and broad economic disruption to impacted areas. Successful detection and early warning for landslides are important for both disaster management and disaster mitigation. In recent decades, technological advancement in areas such as remote sensing and deep learning have been vigorous, and as a result, possibilities for the future of landslide identification include possibilities of effectively monitoring landslides for much less expense than the conventional manual interpretation methods which are extremely time-contained and less precise.

1.1 Motivation

The rising rate and scale of human-caused landslides and global warming require the establishment of strong and scalable detection methods. Early detection of landslide hazard areas will contribute to disaster response planning and mitigation, minimizing possible loss of life and infrastructure damage. Nevertheless, landslide detection is faced with a number of challenges, such as:

- **Data Scarcity:** High-quality labeled datasets for landslide detection are limited, making it difficult to train and validate deep learning models.
- **Class Imbalance:** Landslides occupy only a small fraction of satellite images, leading to imbalanced datasets that bias models toward non-landslide regions.
- **Geographical Variability:** Landslide characteristics vary significantly across different terrains, climates, and geological conditions, making generalization challenging.

1.2 Objective

To study, simulation and performance analysis of state-of-the-art deep learning algorithms for landslide detection using synthetic aperture radar images.

1.3 Outline

The remaining part of this report is organized as follows:

- **Chapter 2 – Background:** A summary of landslide detection approaches with a focus on remote sensing and deep learning with a key point being SAR data advantages.
- **Chapter 3 – Methodology:** Description of the deep-learning models implemented and evaluated in this study.
- **Chapter 4 - Implementation and Training Details:** Description of dataset, evaluation metrics, implementation details and training details
- **Chapter 5 – Results and Discussions:** Presentation and discussion of model performance on various sets of data.
- **Chapter 6 – Conclusion and Future Work:** Summary of results and areas for future work.

Chapter 2

Background

2.1 Methods for Landslide Investigation

2.1.1 Ground Investigation

Ground investigation involves the direct examination and sampling of subsurface materials to obtain precise and localized data. This process is typically executed through a combination of boreholes, trial pits, cone penetration tests (CPT), and other in-situ methods such as the Standard Penetration Test (SPT) and vane shear testing. These investigations yield critical information regarding soil and rock stratigraphy, strength parameters, permeability, groundwater conditions, and potential hazards such as voids or contamination.

These methods support informed decision-making in areas such as foundation design, slope stability analysis, and hazard identification. However, ground investigations are often expensive and time-consuming, particularly in remote or logistically challenging environments. The intrusive nature of these techniques can disturb the natural conditions of the subsurface, potentially altering moisture content or stress distribution. Additionally, because data are collected at discrete locations, spatial extrapolation is limited and may miss critical subsurface variability without a dense sampling framework.

2.1.2 Geological Survey

A geological survey involves the systematic observation, mapping, and interpretation of surface geological features by trained geologists through field-based reconnaissance. Key elements assessed during a survey include lithological composition, stratigraphic sequences, structural features such as faults, folds, joints, and evidence of past geomorphological processes like landslides, erosion, and sedimentation.

These surveys are particularly valuable during the early stages of site assessment, providing essential geological context that can guide more focused investigations. Nevertheless, geological surveys are inherently limited to surface observations and cannot provide detailed information on subsurface conditions. Their effectiveness can be constrained by factors such as dense vegetation, urban development, or poor exposure of rock outcrops. Moreover, the interpretive nature of the work means the quality of results is dependent on the experience and expertise of the geologist, and findings are typically qualitative rather than quantitatively precise.

2.1.3 Remote Sensing Techniques

Remote sensing encompasses a suite of technologies, including aerial photography, satellite imagery, multispectral and hyperspectral sensors, and Light Detection and Ranging (LiDAR). These tools are used to collect data from airborne or spaceborne platforms, facilitating the monitoring of terrain over large spatial extents and extended time periods.

Remote sensing is particularly beneficial for landslide studies and terrain analysis in inaccessible, vast, or hazardous areas where ground access is limited or dangerous. It enables the detection of surface indicators such as cracks, scarps, vegetation stress, and moisture anomalies. Temporal analysis of imagery allows for change detection and the tracking of evolving hazards, enhancing early warning capabilities. However, the utility of remote sensing is constrained by its spatial and spectral resolution. While excellent for regional assessments, it may lack the granularity needed for localized engineering analysis. Data interpretation demands advanced processing skills and the use of specialized software, often requiring multidisciplinary expertise. Additionally, atmospheric interference, such as cloud cover or haze, can degrade image quality, especially in optical sensing systems.



Figure 2.1: Example of NDVI (Normalized Difference Vegetation Index) imagery used in remote sensing to detect vegetation stress and surface changes.

2.1.4 Radar Data

Radar-based methods such as Interferometric Synthetic Aperture Radar (InSAR) detect ground movements by measuring phase differences between radar images. InSAR works by comparing radar images taken at different times from the same location in orbit, detecting minute changes in surface elevation through phase shifts.

One of the main advantages of radar systems is their ability to operate independently of lighting and weather conditions, providing consistent data collection capabilities regardless of time of day or cloud cover. This makes radar particularly useful for long-term monitoring of landslide-prone regions, subsidence areas, and infrastructure movement. It can identify cumulative surface displacements with millimeter-level precision, making it a valuable tool for early warning systems. Nonetheless, radar data comes with inherent limitations. The processing workflow for InSAR is complex and computationally intensive, requiring expertise in signal processing, geodesy, and

geospatial analysis. The accuracy of the results can be affected by temporal decorrelation (due to surface changes between acquisitions), vegetation cover that scatters the radar signal, or steep topography that introduces geometric distortions such as layover and shadowing. Moreover, while InSAR is highly effective for detecting slow, progressive ground movements, it is less suited to capturing rapid, catastrophic events, which may occur between acquisition intervals or be masked by surface disruption.

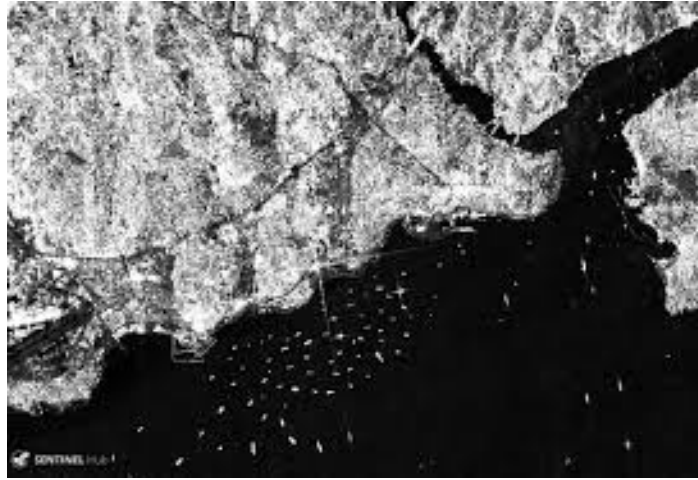


Figure 2.2: Example of Radar image.

2.2 Synthetic Aperture Radar (SAR)

Synthetic Aperture Radar (SAR) is a powerful active remote sensing technique that emits microwave signals and records their return after reflecting off the Earth's surface. SAR systems are mounted on airborne or satellite platforms and provide high-resolution images regardless of weather or lighting conditions, making them especially useful for continuous terrain monitoring in all seasons.

SAR technology forms the basis for advanced techniques like InSAR (Interferometric SAR), which detects surface deformation by measuring phase differences between multiple SAR acquisitions over the same area. InSAR is particularly effective for monitoring slow-moving landslides, subsidence, and other forms of ground instability with millimeter-level precision.

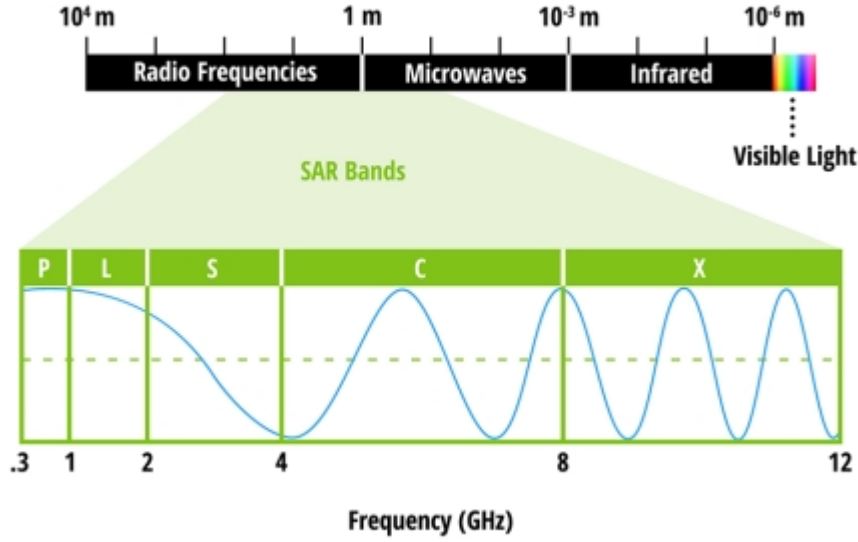


Figure 2.3: SAR Bands (Meyer, 2019)

2.2.1 SAR Polarizations: VV and VH

SAR sensors can transmit and receive signals in different polarization modes, which significantly affect how surface features are detected and classified. Two commonly used polarization configurations are:

- **VV (Vertical Transmit, Vertical Receive):** VV polarization is affected by surface roughness and produces high backscatter for flat, smooth areas such as bare soil or roads. VV polarization is often less variable and less dependent on vegetation cover, so it can be employed to directly assess ground deformation.
- **VH (Vertical Transmit, Horizontal Receive):** VH is a cross-polarization mode most affected by volume scattering, such as vegetation or surface detritus. VH backscatter can be employed to distinguish vegetation cover variations or surface roughness, which can be the earliest landslide activity indication.

The application of VV and VH polarizations can be employed to enhance landslide detection since it can determine the structural deformation (VV) as well as changes on the surface or vegetation disturbance (VH). The VV/VH ratio or difference can also be used in classification routines to separate unstable and stable slopes.

2.2.2 Advantages of SAR over Optical Imagery

SAR sensors transmit microwaves and respond to backscattering of radiative energy in penetrating cloud cover and recovering data independent of lighting. This characteristic is vital to landslide detection in regions of recurrent cloud cover or at night when the application of optical sensors does not ensure the return of reliable data. Furthermore, SAR data can potentially provide ancillary data such as surface roughness and moisture content variation typical of landslide activity. Multitemporal SAR

analysis complements detection reliability by outlining variation evolving in the terrain over time, something that cannot easily be achieved through the application of optical data alone.

Chapter 3

Methodology

3.1 Introduction

This chapter presents the dataset characteristics and the convolutional neural network architectures used for SAR-based landslide segmentation. We first detail the Sentinel-1 derived multi-temporal datacubes and their preprocessing steps. Next, we describe why we selected each model and explore their architectural innovations—focusing on how they address challenges like multi-scale features, attention mechanisms, and computational efficiency in semantic segmentation.

3.2 Deep Learning Models

Accurate segmentation of landslides from SAR imagery presents several challenges: low signal-to-noise ratio, sparse class distributions, complex terrain topography, and variable surface backscatter. To address these issues, we explored a diverse range of deep learning architectures, each bringing unique strengths in handling spatial detail, scale variation, and multimodal fusion. Below, we detail each model’s motivation, architecture, and relevance to our task.

3.2.1 U-Net

U-Net was introduced for biomedical image segmentation (Ronneberger *et al.* (2015)), where fine spatial resolution and limited training data are common. It features a symmetric encoder–decoder structure with skip connections that bridge encoder features to corresponding decoder layers. This allows precise reconstruction of object boundaries from high-level semantic features. UNet was selected for its effectiveness in handling small and sparse targets characteristics that align well with landslide segmentation. The skip connections are particularly valuable for preserving boundary detail, which is essential for delineating narrow and irregular landslide patterns. Its express simplicity and reasonable performance under sparse data qualify it as a reasonable baseline for SAR-based segmentation.

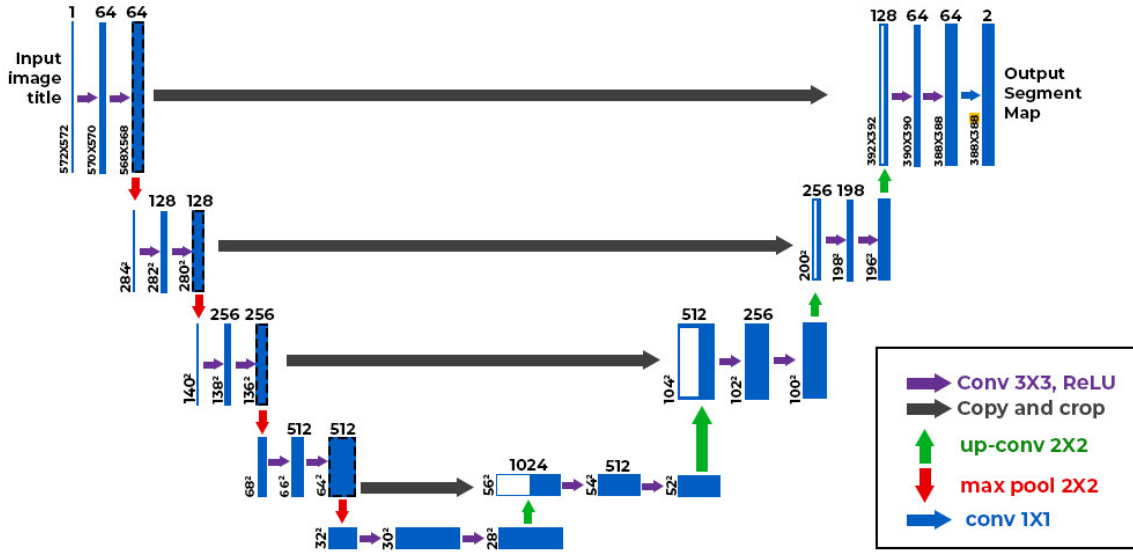


Figure 3.1: U-Net Architecture

3.2.2 LinkNet

LinkNet is a lightweight encoder-decoder framework constructed based on ResNet-like blocks (Chaurasia and Culurciello (2017)). It lowers the number of parameters without compromising semantic richness by employing residual skip connections. Pioneered in the use of real-time applications in the field of segmentation, LinkNet works best in time-sensitive applications of disaster response. Its framework of residual learning supports improved gradient flow and the retention of relevant semantic context. For SAR imagery, where computational effectiveness and robustness are essential, LinkNet balances model simplicity with accuracy in the field of segmentation.

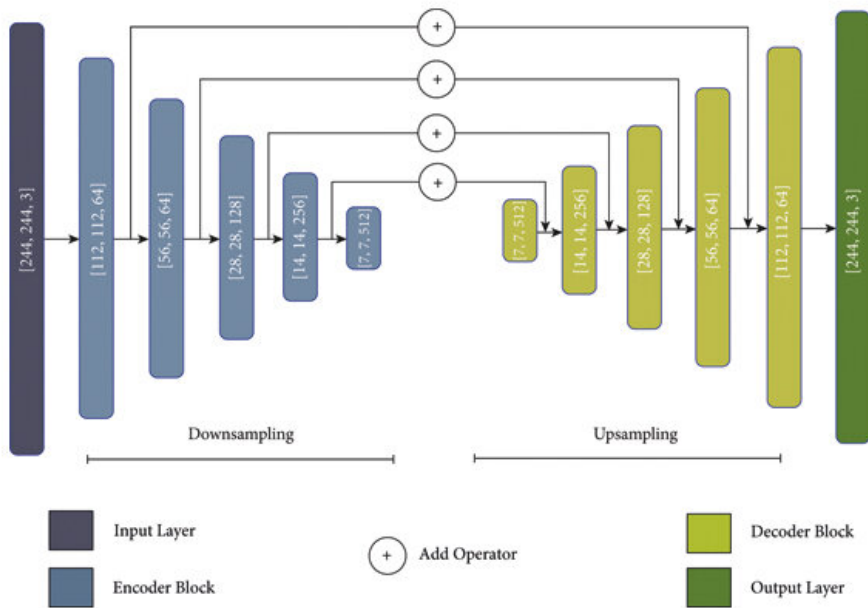


Figure 3.2: LinkNet Architecture

3.2.3 U-Net++

U-Net++ introduces a more sophisticated skip connection strategy with dense convolutional blocks between encoder and decoder stages, improving semantic alignment across resolutions (Zhou *et al.* (2018)). The nested and dense architecture in UNet++ enables better fusion of low-level spatial and high-level semantic information. This is particularly beneficial in landslide detection, where subtle changes in terrain and surface texture must be accurately segmented. UNet++ performs well on small or fragmented regions, making it suitable for complex landslide geometries often seen in SAR data.

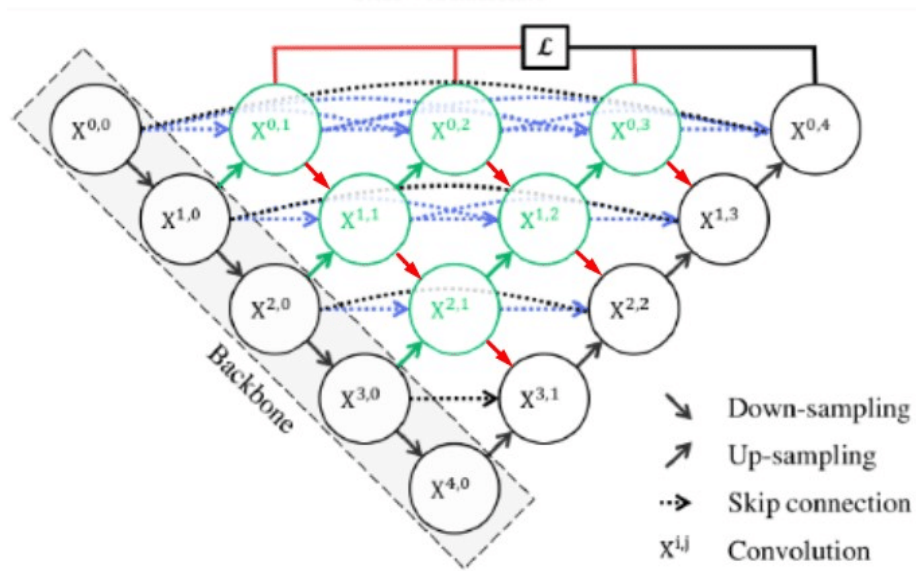


Figure 3.3: UNet++ Architecture

3.2.4 Pyramid Attention Network (PAN)

PAN incorporates spatial attention mechanisms, along with pyramid pooling, into an encoder-decoder network (Li *et al.* (2018)), enabling segmentation that is aware of context and relevant over a large receptive field. Because it combines attention with multi-scale pooling, PAN can keep track of important context and important features in the image. This capability is useful when working with SAR data that may have substantial background clutter and noise in the imagery. Therefore, PAN's capacity to emphasize informative spatial structures enhances the robustness of segmentation in different terrain and varying scales of landslide.

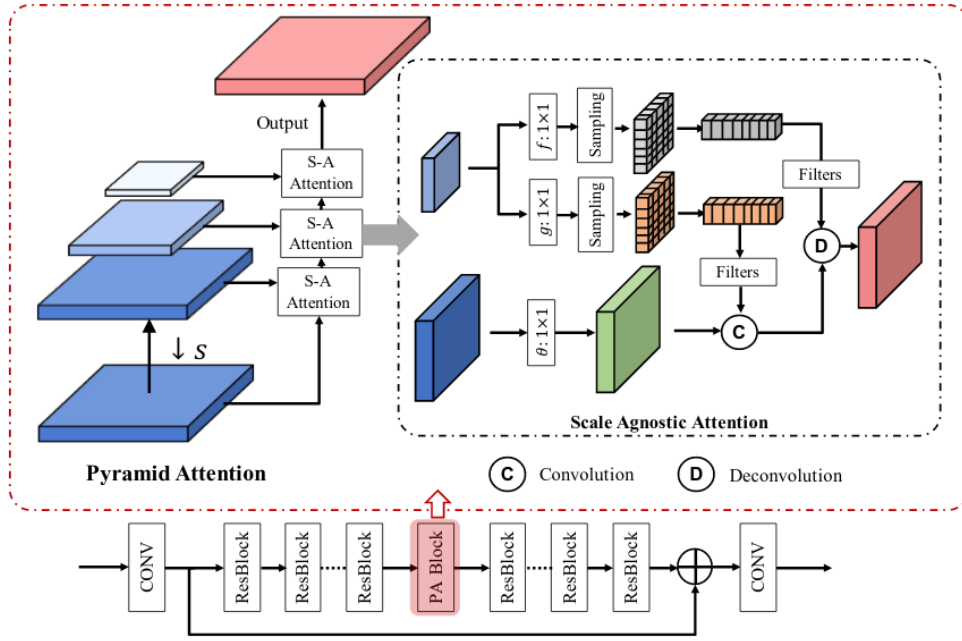


Figure 3.4: Pyramid Attention Network (PAN) Architecture

3.2.5 DeepLab v3+

DeepLab v3+ is based on Atrous Spatial Pyramid Pooling (ASPP) and captures features across multiple scales, addressing challenges with scale variations of the landslide and fragmentation of individual landslide objects (Chen *et al.* (2018)). By applying atrous convolutions that utilize different dilation rates, the model is able to integrate both fine and coarse contextual information. This integration is necessary when investigating landslides that occur and manifest across differing spatial extents. In addition to segmentation ability, DeepLab v3+ decodes image features to refine object boundaries, as well as provide relevant information specifically useful for SAR imagery and high-resolution segmentation tasks.

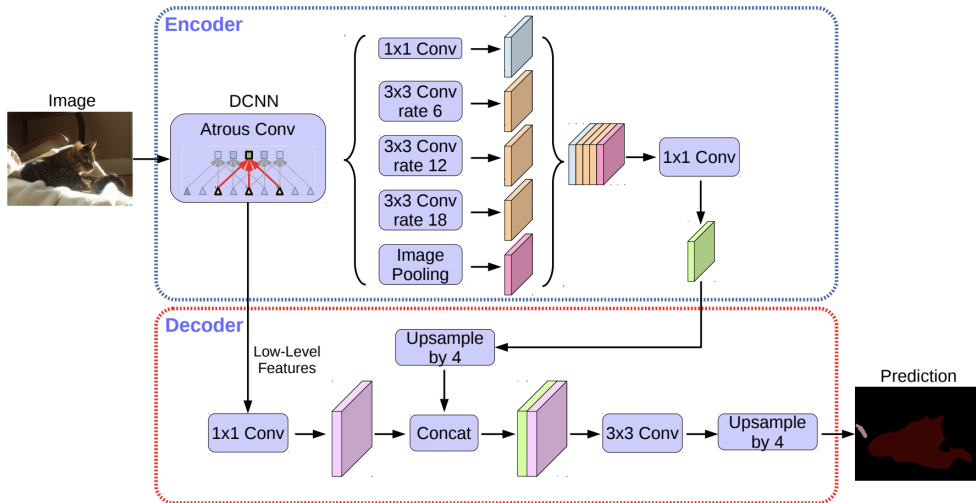


Figure 3.5: DeepLab v3+ Architecture

3.2.6 DRs-UNet

DRs-UNet modifies the standard UNet architecture by embedding Dense Residual (DR) blocks within the architecture, combining DenseNet’s strengths of feature reuse and ResNet’s gradient stability (Chen *et al.* (2022)). This merge of architectures directly relates to improvements in representation learning and the mitigation of vanishing gradients, allowing the networks to go deeper while retaining spatial information. For SAR-based landslide detection, DRs-UNet can effectively capture subtle changes in surface structure and backscatter, making it well-suited for detection of landslides in an early stage of development or thin-surface landslides.

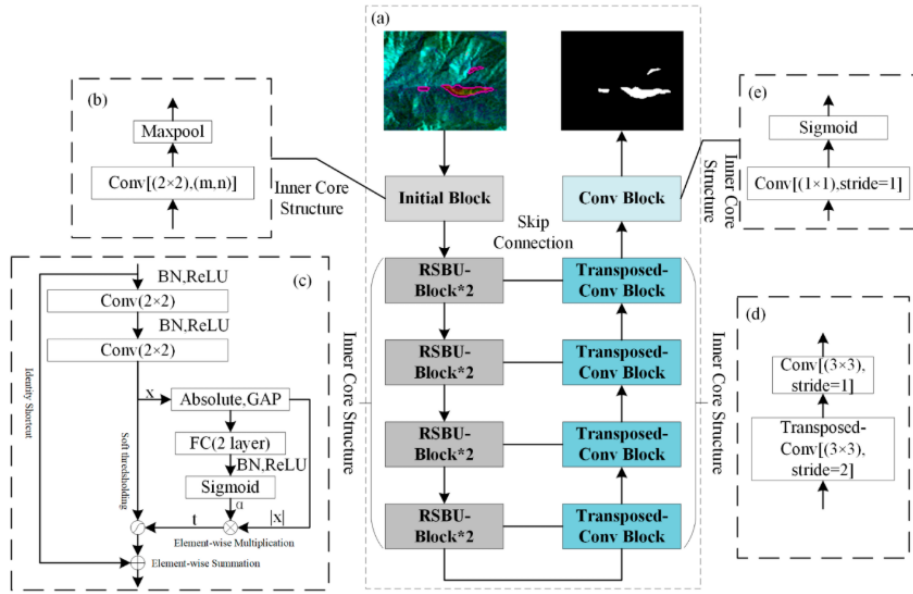


Figure 7. Schematic of the proposed DRs-UNet: (a) overall structure of DRs-UNet; (b) initial block; (c) inner core structure of RSBU-block; (d,e) inner core structure of transposed convolution block (TCB) and the final convolution block.

Figure 3.6: DRs-UNet Architecture

3.2.7 MSSCSAF-Net

MSSCSAF-Net employs a Multiscale Skip-connected Channel-Spatial Attention Fusion method (El Rai *et al.* (2024)), which can learn both inter-channel associations and spatial contextualizations essential for landslide shape recognition, especially when integrated together. The model is particularly advantageous in dealing with multiple input modalities such as VV, VH and DEM, since attention is learned to these relevant spectral (VV, VH,) and spatial (DEM) cues (which can sometimes be relevant to geolocation) in the data. Its multiscale design facilitates detection at various spatial resolutions, which is important in SAR imagery, since where it was collected will determine the degree of spatial complexity that simply does not exist in optical imagery. MSSCSAF-Net is highly suitable for scenarios demanding fine segmentation across multimodal data sources.

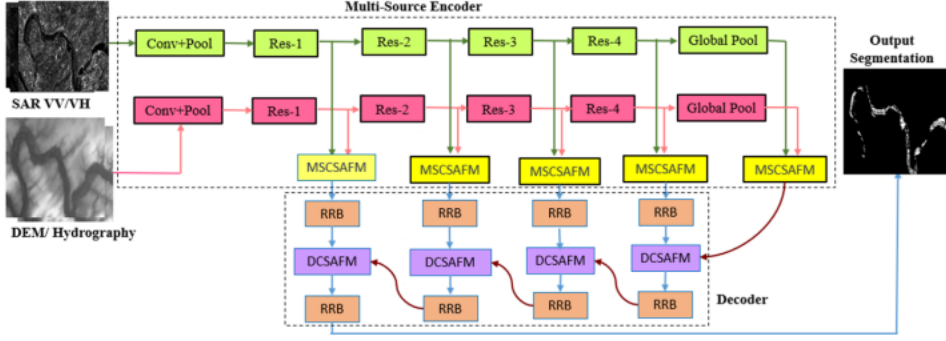


Figure 3.7: MSSCSAF-Net Architecture

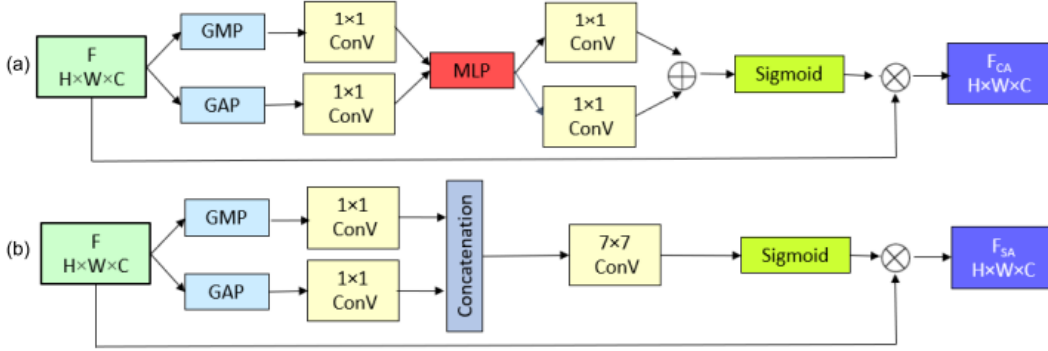


Figure 3.8: (a) Channel Attention (b) Spatial Attention Module

3.2.8 ASKU-Net++

ASKU-Net++ is an advanced attention-based model that adapts UNet++ with asymmetric convolutions and skip connections tailored to SAR-specific noise patterns (Tang *et al.* (2025)). This approach increases orientation sensitivity and suppress noise which is important in SAR segmentation processes. Asymmetric convolutions and attention gates enable the model to capture directional and elongated aspects in its prediction (i.e. landslides on slopes). Its architecture has been specifically designed to deal with the type of noise and geometric distortion that exists within SAR images, allowing it to be strong candidate architecture for applications in the real-world where the SAR data is largely affected by both noise levels and geometric distortion irrespective of the intention of the satellite for its operational models.

Chapter 4

Implementation and Training Details

4.1 Dataset Description

The data used in this research are multi-temporal SAR datacubes constructed a per the method stated in Boehm *et al.* (2022), specifically for the purpose of landslide segmentation and detection applications, with respect to the datacube, was built with data from the European Space Agency (ESA) controlled Sentinel-1 satellite mission. The Sentinel-1 satellite mission gathers C-band, double-polarized (VV and VH) SAR imagery in Interferometric Wide (IW) swath mode which is provides an optimum balance between reasonable spatial resolution and temporal repeat frequency for addressing natural disasters.

4.1.1 Dataset-I: 2017 Hokkaido, Japan

The 2017 landslides in Hokkaido were primarily triggered by a sequence of seismic and meteorological events. A moderate earthquake in early 2017 weakened slope stability across the region, and this was followed by intense and prolonged rainfall during the summer months. These compounding factors caused numerous shallow and deep-seated landslides, particularly in mountainous and forested areas with volcanic ash-rich soils.

These landslides disrupted local infrastructure, blocked roads, and threatened rural communities. The geology of Hokkaido, with steep slopes and loose volcanic deposits, contributed to widespread mass movement following saturation. The event is considered an important case study for understanding multi-trigger landslide dynamics and for testing early warning and detection systems.

Dataset Overview

- 26 temporal acquisitions from Sentinel-1,
- Dual-polarized SAR intensity imagery (VV and VH),
- Scene coverage of 5382×4398 pixels,
- Aligned Digital Elevation Model (DEM) data,

- Manually selected binary landslide segmentation masks with pixel-level annotations.

This dataset captures how surface deformation and SAR backscatter signatures evolved over time, enabling detailed spatiotemporal analysis of landslide onset and progression.

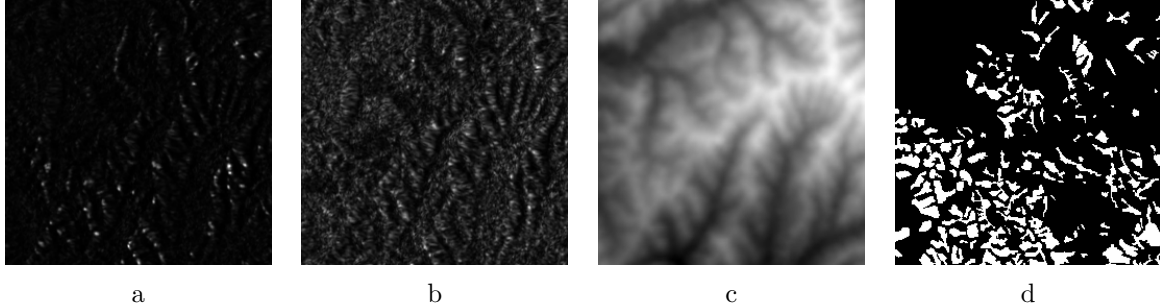


Figure 4.1: (a) VV (b) VH (c) DEM (d) Label
Sample Patch from Dataset-I

Table 4.1: Class distribution in Dataset-I

Class	Percentage of Total Pixels
Landslide (Positive)	1.9802 %
Non-Landslide (Negative)	98.0198 %

4.1.2 Dataset-II: Mt. Talakmau, Indonesia

In early 2022, a strong earthquake struck the region around Mt. Talakmau, located in West Sumatra, Indonesia. The seismic event triggered widespread landslides along the volcanic slopes of the mountain. These landslides were further exacerbated by heavy rainfall in the days following the quake, creating conditions ripe for slope failures in saturated, weathered volcanic soils.

The affected areas include densely forested and agricultural regions with high topographic relief. Landslides led to significant ecological disruption, damage to agricultural lands, and temporary displacement of nearby communities. The terrain and tropical vegetation introduced additional complexity for remote sensing-based detection and analysis.

Dataset Overview

- 20 temporal Sentinel-1 SAR acquisitions,
- Same VV and VH polarization channels,
- Aligned Digital Elevation Model (DEM) data,
- Spatial dimensions of 4313×3006 pixels.

This dataset provides a contrasting landscape and climate, useful for assessing model generalization to varied terrain, vegetation density, and climatic triggers.

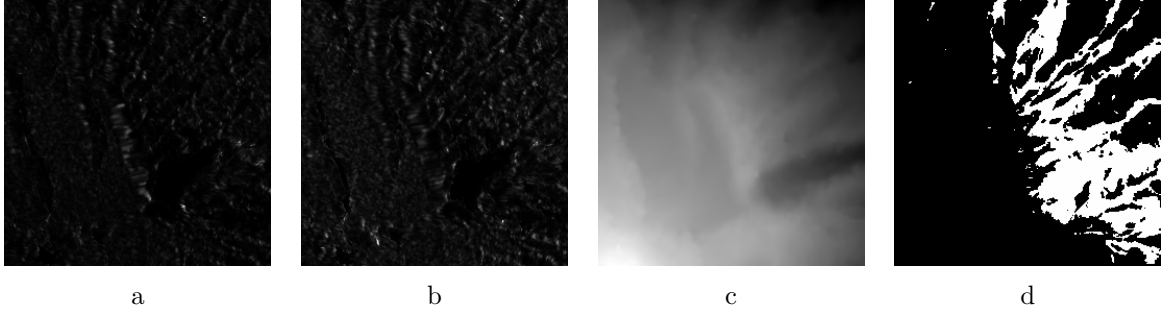


Figure 4.2: (a) VV (b) VH (c) DEM (d) Label
Sample Patch from Dataset-II

Table 4.2: Class distribution in Dataset-II

Class	Percentage of Total Pixels
Landslide (Positive)	0.4801 %
Non-Landslide (Negative)	99.5199 %

4.2 Evaluation Metrics

To accurately assess the segmentation performance of deep learning models on SAR-based landslide detection, both threshold-independent and threshold-dependent metrics were used. These metrics give insight into the model’s ability to correctly identify landslide regions, even when most of the background consists of non-landslide areas.

4.2.1 Precision

Precision (Powers (2011)) refers to the proportion of the model’s positive predictions that are actually correct. It highlights the model’s ability to minimize false positives, which is critical in applications like landslide detection, where false alarms can lead to unnecessary interventions or incorrect risk assessments.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.1)$$

In SAR-based landslide segmentation, high precision means the model is cautious in labeling regions as landslides, reducing overestimation. However, this needs to be balanced with recall to avoid missing true landslides.

4.2.2 Recall

Recall (Powers (2011)), also called sensitivity or the true positive rate (TPR), measures the model’s ability to correctly identify all actual positive instances. High recall indicates the model is accurately predicting most landslide pixels present in the ground truth.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.2)$$

This is particularly important for early warning systems, as missing a real landslide

(false negative) could have serious consequences. Recall is often prioritized in safety-critical systems, although it may increase false positives if not balanced with precision.

4.2.3 F1 Score

The F1 score (Powers (2011)) provides a harmonic mean of precision and recall, yielding a single scalar value that captures both sides of the performance metric. It is defined as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.3)$$

The F1 score balances precision and recall while also penalizing for extreme imbalances between precision and recall. Thus, it is an appropriate measure for being able to determine how well the model distinguishes landslide boundaries from the surrounding terrain. IoU indicates how much overlap the predicted region has with the ground truth.

4.2.4 IoU

Intersection over Union (IoU) (Rezatofighi *et al.* (2019)), also known as the Jaccard Index, is a measure of overlap between predicted and ground truth masks. It is widely used for semantic segmentation tasks especially when distinguishing precise boundary delineation is critical.

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (4.4)$$

IoU penalizes both over-segmentation (false positives) and under-segmentation (false negatives) which together it provides a more well-rounded performance metric. In terms of landslide detection using SAR imagery, a high IoU indicates that the predicted landslide area closely resembles the actual landslide that is observed in the ground truth which is critically important for accurate mapping and disaster response.

4.2.5 Area Under Precision-Recall Curve (AUPRC)

AUPRC (Saito and Rehmsmeier (2015)) quantifies the relationship between precision and recall at different decision thresholds. AUPRC is not considered a perfect substitute for the Receiver Operating Characteristic (ROC) curve but is more suitable for class-imbalance problems compared to evaluating traditional ROC curves (e.g., landslide detection using SAR imagery). The sparse occurrence of positive pixels (landslides) in the imagery (compared to negative pixels or non-landslide areas) causes a class imbalance issue in this situation.

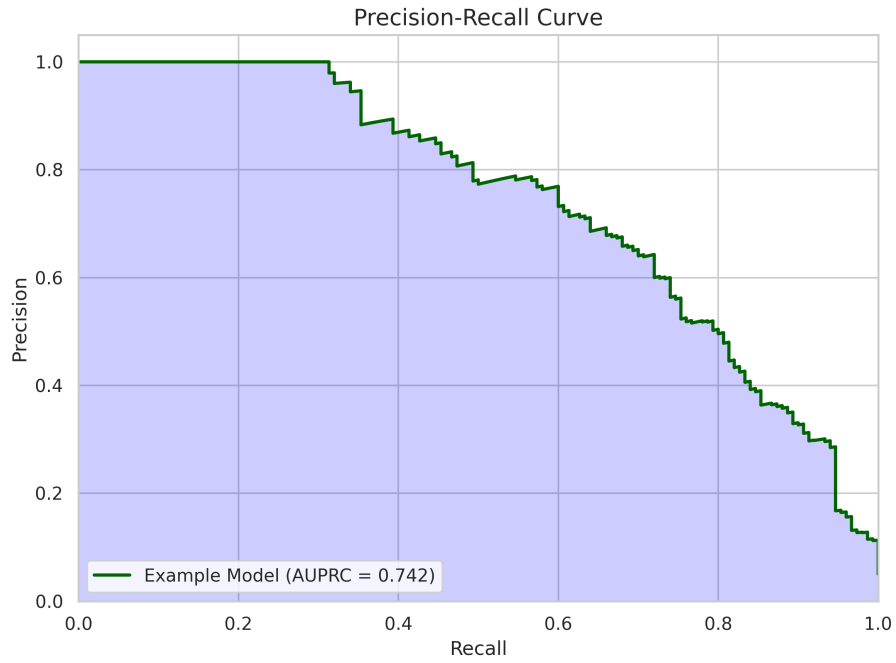


Figure 4.3: Example Precision-Recall

4.2.6 Confusion Matrix

		Predicted Values	
		Positive	Negative
Actual Values	Positive	TP	FN
	Negative	FP	TN

Figure 4.4: Confusion Matrix

- where:
- TP : True Positives
 - FP : False Positives
 - FN : False Negatives
 - TN : True Negatives

4.3 Implementation Details

4.3.1 Dice Loss

Dice Loss (Milletari *et al.* (2016)) is a differentiable function based on the Sørensen–Dice coefficient, commonly used to measure the similarity between two samples. It’s especially useful for training neural networks on imbalanced segmentation tasks, such as landslide detection, where the positive pixels are much fewer than the negative ones. The Dice coefficient (D) is calculated over all pixels in the image or patch,

The Dice coefficient D is defined as:

$$D = \frac{2 \cdot |P \cap G|}{|P| + |G|} = \frac{2 \sum_i p_i g_i}{\sum_i p_i + \sum_i g_i} \quad (4.5)$$

Where:

- p_i is the predicted probability for pixel i ,
- g_i is the ground truth label for pixel i ,
- The summation is over all pixels in the image or patch.

and the Dice Loss is computed as:

$$\mathcal{L}_{\text{Dice}} = 1 - D \quad (4.6)$$

This formulation encourages overlap between prediction and ground truth and penalizes both false positives and false negatives equally. Unlike pixel-wise losses such as binary cross-entropy, Dice Loss captures spatial coherence, which is crucial in delineating the often thin and irregular geometries of landslides.

4.3.2 Optimizer

The AdamW optimizer was utilized, set with a learning rate of 1×10^{-2} . and a weight decay of 1×10^{-4} . AdamW was chosen for its decoupled weight decay feature, which improves generalization by addressing the shortcomings of traditional L2 regularization within adaptive gradient methods. The use of this particular optimizer, along with consistent hardware and data settings, facilitated stable and reproducible training performance through each experimental run.

4.4 Training Details

To ensure compatibility with convolutional neural network architectures and enable efficient mini-batch processing, the original datasets were spatially cropped such that both horizontal (x) and vertical (y) dimensions are multiples of 256 pixels. This pre-processing step allowed for seamless extraction of uniformly sized image tiles. From the preprocessed Japan datacube, patches of size 256×256 pixels were extracted to form the training and testing datasets. Each sample comprised data from twelve temporal acquisitions, including both VV and VH polarization channels. An identical

preprocessing pipeline was applied to the Indonesian dataset, resulting in a structurally equivalent validation set. This setup enabled cross-regional generalization assessment.

All experiments were conducted on the Kaggle platform utilizing an NVIDIA Tesla P100 GPU. The model was trained for 100 epochs with a batch size of 4, using an 80% training and 20% testing data split.

Chapter 5

Results and Discussions

5.1 Introduction

This chapter provides a thorough analysis of various state-of-the-art deep learning models for landslide segmentation with SAR imagery. These models were analyzed using two distinctive datasets that exhibited different strategies for data sampling—high-coverage and low-redundancy sampling—to determine how the two different levels of sampling impacted segmentation performance. The chapter provides quantitative results, qualitative visualizations, and a comparative analysis of the high-coverage sampling datasets on the primary dataset. In addition, the best models were tested using a second distinctive dataset from an entirely different geographic location to determine their performance and generalizability. The study provided insight into the ability of various architectures, and sampling methods, to build an effective SAR-based landslide detection system.

5.2 Benchmark Model Evaluation on Dataset-I

5.2.1 Benchmark Model Performances Using High-Coverage Sampling

This subsection evaluates the segmentation performance of various models trained using a high-coverage sampling strategy, which ensures broad spatial representation and class balance in the training data.

Table 5.1: Benchmark model performances using High-Coverage Sampling:

Model	Parameters	Precision	Recall	AUPRC	F1 Score	IoU
UNet (2015)	5.1	0.840	0.823	0.855	0.831	0.711
LinkNet (2017)	5.1	0.800	0.789	0.815	0.794	0.665
UNet++ (2018)	16.1	0.835	0.821	0.869	0.828	0.704
PAN (2018)	2.4	0.845	0.832	0.847	0.838	0.720
DeepLabv3+ (2018)	17.6	0.890	0.876	0.891	0.883	0.770
DRs-UNet (2022)	8.1	0.821	0.799	0.834	0.811	0.686
MSSCSAF-Net (2024)	19.0	0.910	0.898	0.887	0.904	0.792
ASKU-Net++ (2025)	10.7	0.895	0.878	0.884	0.897	0.773

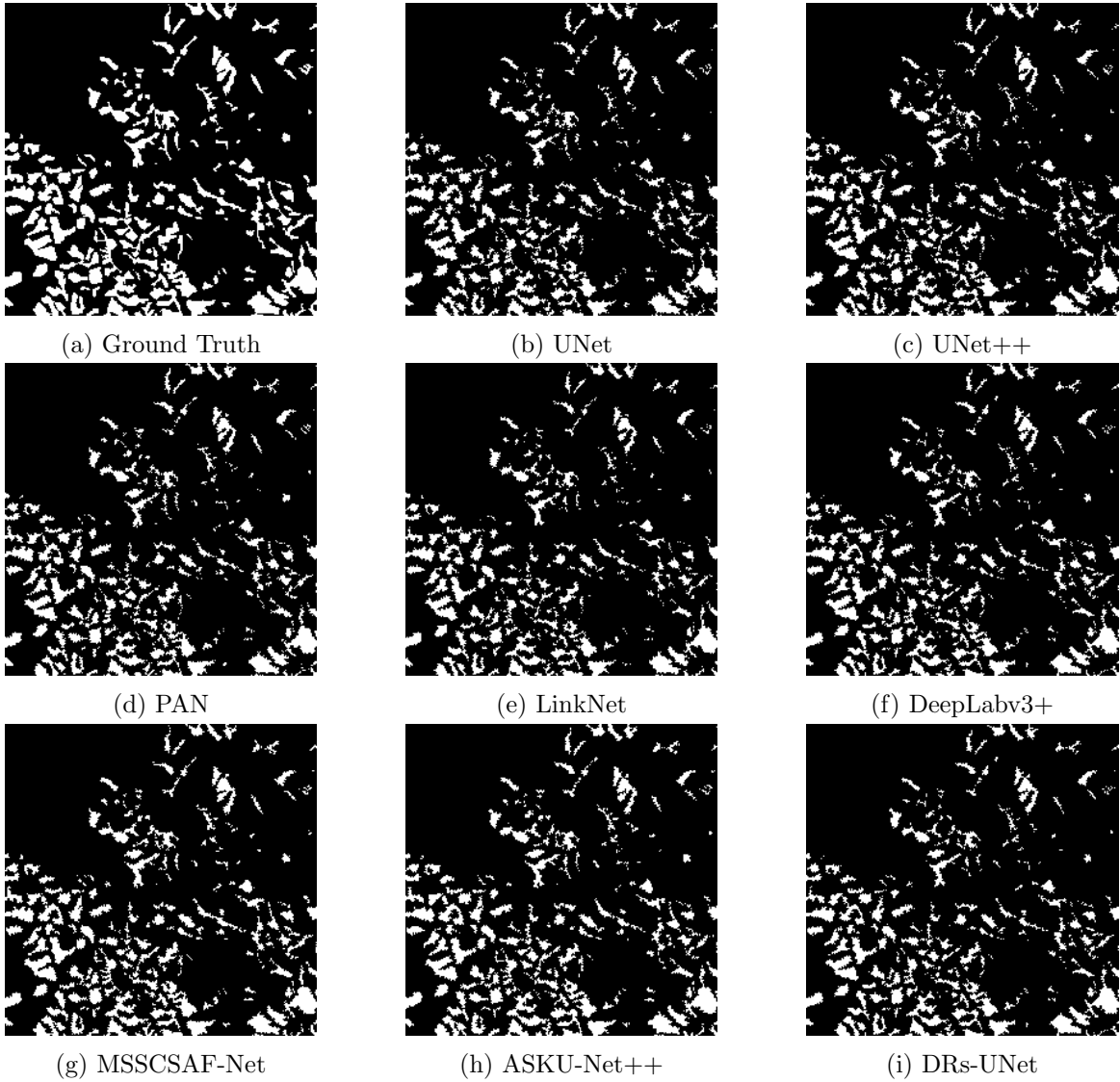


Figure 5.1: Ground truth label and model predictions using High-Coverage Sampling.

5.2.2 Benchmark Model Performances Using Low-Redundancy Sampling

This subsection analyzes model performances when trained with a low-redundancy sampling strategy, aimed at reducing data duplication while maintaining spatial diversity and representational efficiency.

Table 5.2: Benchmark model performances using Low-Redundancy Sampling:

Model	Parameters	Precision	Recall	AUPRC	F1 Score	IoU
UNet (2015)	5.3	0.823	0.801	0.832	0.812	0.689
LinkNet (2017)	4.8	0.790	0.766	0.793	0.778	0.635
UNet++ (2018)	8.0	0.820	0.797	0.841	0.808	0.685
PAN (2018)	1.9	0.822	0.800	0.820	0.811	0.688
DeepLabv3+ (2018)	17.3	0.850	0.833	0.857	0.841	0.727
DRs-UNet (2022)	8.1	0.805	0.765	0.812	0.785	0.654
MSSCSAF-Net (2024)	18.8	0.860	0.845	0.873	0.852	0.736
ASKU-Net++ (2025)	10.7	0.870	0.834	0.856	0.851	0.731

Qualitative Comparison

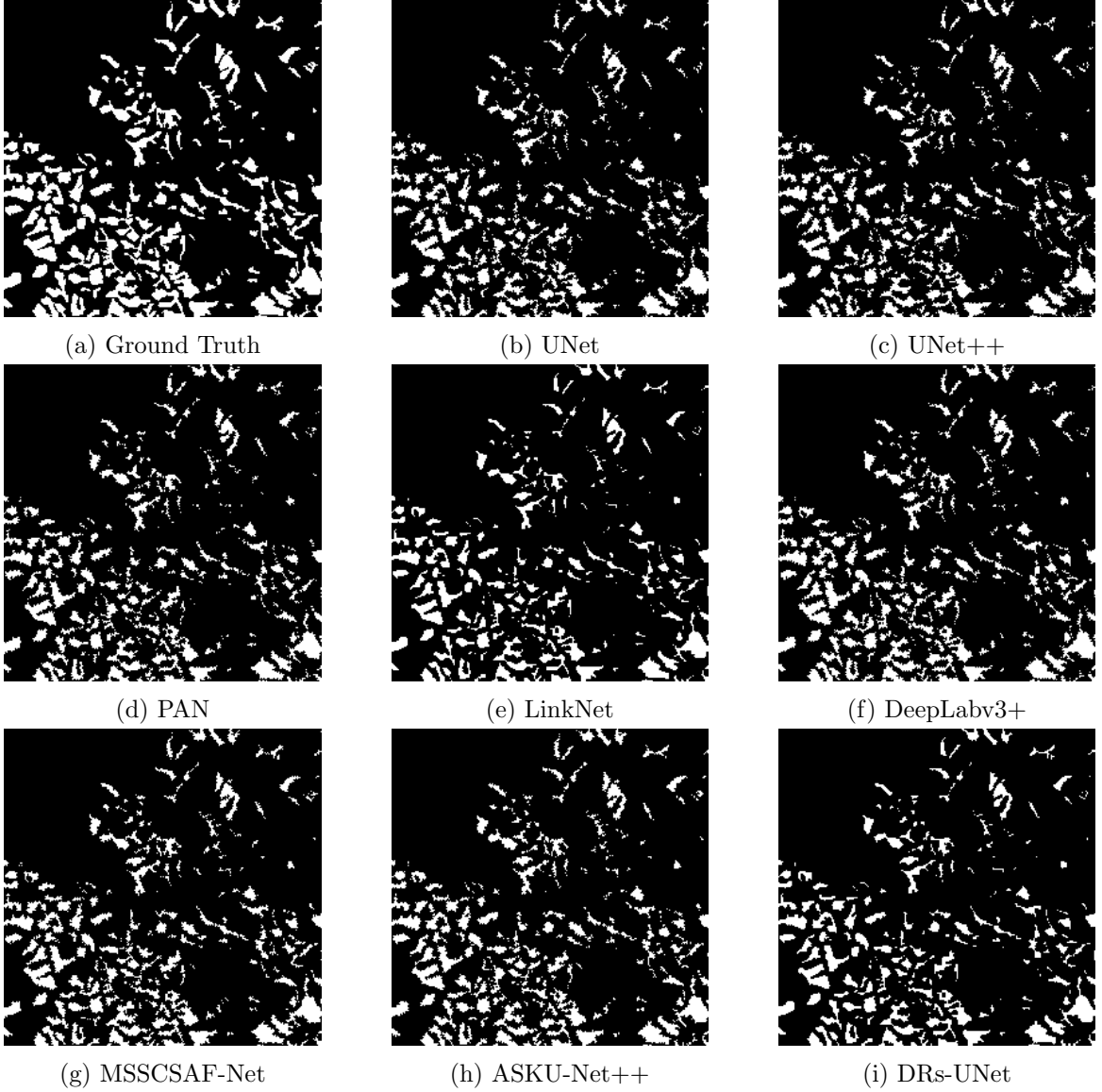


Figure 5.2: Ground truth label and benchmark model predictions using Low-Redundancy Sampling.

5.2.3 Comparative Analysis

From the results, several key observations emerge:

- Approach 1’s high-coverage sampling scheme usually attained higher segmentation performance, which could be ascribed to class-balanced representation and higher spatial diversity in the training set.
- Despite less redundancy, Approach 2 (low-redundancy sampling scheme) still provided competitive performance, and notably, for MSSCSAF-Net, which preserved the best performance through much less duplicated data.
- MSSCSAF-Net always surpassed all the other models in F1 score for both sampling strategies, proving effective in making use of multi-scale and multi-modal information for accurate landslide segmentation.
- DeepLabv3+ likewise ran well, obtaining the best AUPRC under the high-coverage condition, and was still a contender under the low-redundancy scenario.
- PAN demonstrated excellent performance considering its light architecture, once again proving that an effective model can be competitive when accompanied by proper sampling and regularization methods.

These findings underscore the critical role of spatial sampling strategies in designing effective SAR-based landslide segmentation pipelines.

5.3 Benchmark Model Evaluation on Dataset-II

5.3.1 Evaluation Setup

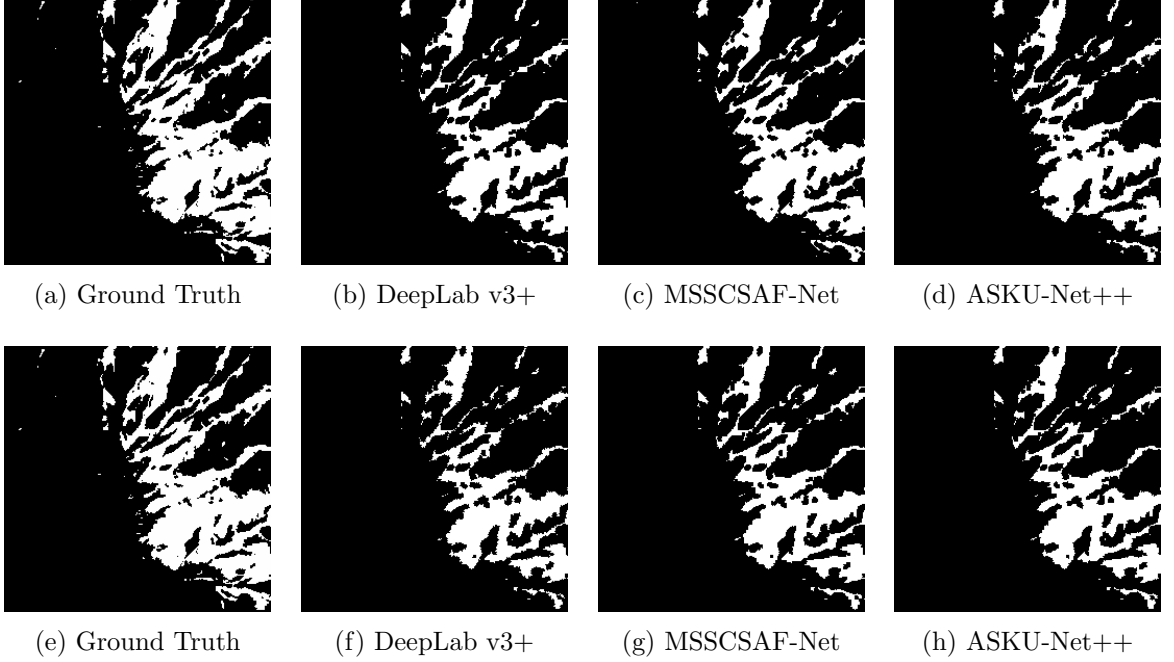
To further investigate the high robustness and cross-domain generalizability of the top-performing models, we accepted the challenge to test ASKU-Net++, MSSCSAF-Net and DeepLabV3+ the top-performing models under both high-coverage and low-redundancy data sampling strategy scenarios—on the auxiliary SAR-based landslide dataset for Mt. Talakmau, Indonesia. We can, in this way, approximate the generalizability of models trained under different data sampling schemes in a new geographic region exhibiting different terrain, coverage, and natural environment.

5.3.2 Benchmark Model Performances on Dataset-II

Table 5.3: Performances of top models from each sampling strategy on the Dataset-II.

Sampling Strategy	Model	AUPRC	F1 Score	Precision	Recall	IoU
High-Coverage	DeepLabV3+	0.761	0.747	0.765	0.730	0.678
	MSSCSAF-Net	0.805	0.786	0.812	0.761	0.726
	ASKU-Net++	0.803	0.790	0.798	0.782	0.703
Low-Redundancy	DeepLabV3+	0.741	0.707	0.730	0.685	0.642
	MSSCSAF-Net	0.781	0.753	0.774	0.735	0.695
	ASKU-Net++	0.766	0.757	0.762	0.752	0.670

5.3.3 Qualitative Comparison



Segmentation results on Dataset-II. Top row: High-Coverage sampling. Bottom row: Low-Redundancy sampling.

5.3.4 Analysis and Observations

MSSCSAF-Net consistently achieved the highest F1 scores on the secondary dataset across both sampling strategies, demonstrating strong generalization and robustness to domain shifts.

DeepLabV3+ and ASKUNET++ both exhibited competitive performance, both closely trailing MSSCSAF-Net. This highlights their capability to maintain robustness and effectively capture complex segmentation features from SAR and terrain data.

The gap in performance between high-coverage, low-redundancy strategies on the secondary set was relatively narrow, indicating both methods yield transferable representations with little loss in out-of-domain performance.

Models trained with the high-coverage sampling strategy generally exhibited slightly better generalization, likely due to improved boundary supervision and more balanced class distributions during training.

This comparative evaluation highlights the point that even when high sampling strategies all optimize in-domain performance, in fact, superior, low-redundancy sampling strategies are able to offer models that are able to attain good generalization performance.

Notably, MSSCSAF-Net was the most stable and high-performance configuration across all the sampling strategies as well as all the geographic regions, and therefore is a strong contender for use in SAR-based landslide detection applications.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This study evaluated advanced deep learning models for landslide segmentation using SAR imagery, under two data sampling strategies: high-coverage and low-redundancy. Results showed that MSSCSAF-Net consistently outperformed other models in both in-domain and cross-domain settings, demonstrating strong generalization and robustness. DeepLabV3+ and ASKU-Net++ also performed well, particularly with high-coverage sampling.

High-coverage sampling generally led to better in-domain performance due to greater spatial diversity and class balance. However, low-redundancy sampling offered comparable results with reduced data duplication, making it an efficient alternative. Cross-domain tests on a secondary dataset confirmed that both strategies can produce transferable models when paired with strong architectures. Overall, MSSCSAF-Net emerged as the most reliable model across all conditions, making it a strong candidate for practical SAR-based landslide detection applications.

6.2 Future Scope

Exploring transformer-based architectures can enhance segmentation accuracy, improving model performance. Additionally, evaluating these models on varied SAR datasets from different regions and conditions ensures robustness and supports broader applicability.

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Appendix

RESUME

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